Table 5. Global clinical, translational, and resource programs for salivary gland organoid research and therapy

Program / institution	Approach / indication	Cell product or resource / delivery	Stage and key findings / status	Refs
RESTART / University Medical Center Groningen (Netherlands)	Autologous salivary organoid-derived cell transplantation for radiation-associated salivary dysfunction after oral cavity cancer surgery and postoperative (chemo)radiotherapy.	Patient-derived cells expanded ex vivo. Historical participant information describes injection into the remaining irradiated submandibular gland about 4 weeks after radiotherapy; the route/timing are planned, not proof of treatment delivery.	Phase I, open-label, single-group trial (NCT04593589; estimated n = 18; one centre). Systemic diseases affecting salivary function (including SjD) are excluded. Primary evaluation: safety and salivary-flow feasibility. Published manufacturing includes a 4-hour final-product handling window at 5 ± 3 °C. No posted clinical results as of 22 October 2026.	22; N2; N3
ATORM-S / OrganoidSciences (South Korea)	Autologous salivary organoid regenerative therapy for refractory gland hypofunction / xerostomia.	Patient-derived salivary gland organoid platform; clinical product specification, dose and delivery details are not established in peer-reviewed clinical reports.	Company-disclosed development program. Related salivary organoid transplantation research is preclinical; company plans do not establish trial initiation or human efficacy.	77
Mayo Clinic salivary tissue–organoid biorepository (USA)	Clinically annotated research resource supporting regenerative medicine and disease modeling; not a therapeutic intervention.	Salivary tissue and organoid biorepository established from 208 donors; used for progenitor characterization and assay development.	Established biorepository resource. Supports future translational research but does not provide transplantation safety or efficacy outcomes.	23

RESTART registration and treatment details describe a prospective trial and its published manufacturing process; they should not be interpreted as evidence of clinical efficacy. The 5×10^4–5×10^6 cells/vial range is a manufacturing specification, not a verified administered dose.

ATORM-S development stage is based on company disclosures, whereas transplantation efficacy data cited for this program remain preclinical. References shown here retain the current manuscript/source identifiers and will be reconciled with the final bibliography.

Abbreviations: SG, salivary gland; GMP, good manufacturing practice; SjD, Sjögren's disease; NCT, ClinicalTrials.gov identifier.